

Designing With Protection Against EMC, EMI And ESD

The effects of electrostatic discharge and interference is often ignored by designers, which results in their rejection, especially in global markets. The designers need to take care of such international standards as FCC, CE, and VCCI if they want their products to succeed in international markets. This article explains the causes of failure as well as the remedies

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EMI stands for electromagnetic interference, EMC for electromagnetic compliance, and ESD for electrostatic discharge. These acronyms define how a product can unintentionally become a source of radio frequency waves generation, or how it can become a victim of radio frequency waves that come from various sources in the environment.

ESD defines how robust a product is to withstand the static voltage discharge that is accumulated by the human operator. Standard bodies like FCC (Federal Communication Commission) in the USA, CE (Conformité Européenne) in Europe, and VCCI (Voluntary Control Council for Interference) in Japan specify the amount of radiation an electronic product can radiate, or the amount of radiation an electronic product should withstand without any

permanent failure. Similarly, there are voltage discharge levels that a product should withstand without getting damaged.

With increased use of electronics in our daily life, compliance to these standards ensures that the users are safe and products do not pollute and affect the working of products in the vicinity. While each country has its own standards, they do not vary much in values. However, from the electronic packaging point of view we need to know what standards the products should meet. Knowing the standards helps in designing the packaging such that the products meet the required standards. The best strategy is to get the worst-case limits and design the product for them. However, cost of doing this may be high in some cases.

Let us understand the basics of the radiation mechanism for designing a compliant product. Almost all electronic products radiate or get impacted in either of the two modes. These are:

Conducted emission (CE). Power and other cables act as carrier and radiate unwanted RF signals.

Radiated emission (RE). Product itself radiates, like an RF transmitter.

The frequency limits for conducted and radiated emissions are different. Most emissions are radiated emissions, and these are spread over large bandwidths.

Radiated emission requires an RF transmitter

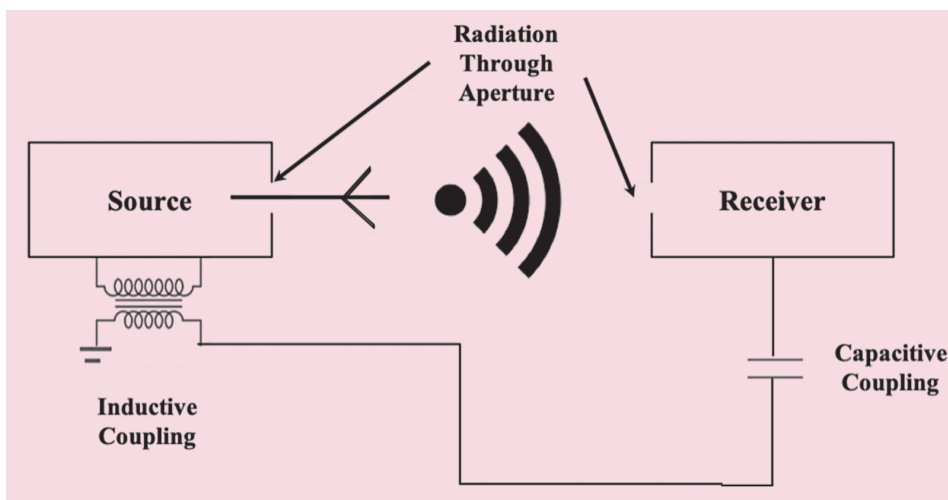


Fig. 1: Radiated and conducted emission mechanism in a product